

Agentic AI

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AI has evolved in both capabilities and autonomy

**“Dumb”
AI**

Hard-coded
automation



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**Predictive
AI**

Hard-coded
automation

Pattern recognition
Outcome prediction



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Knowledge overview
Content generation



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Knowledge overview
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Complex reasoning
Task execution



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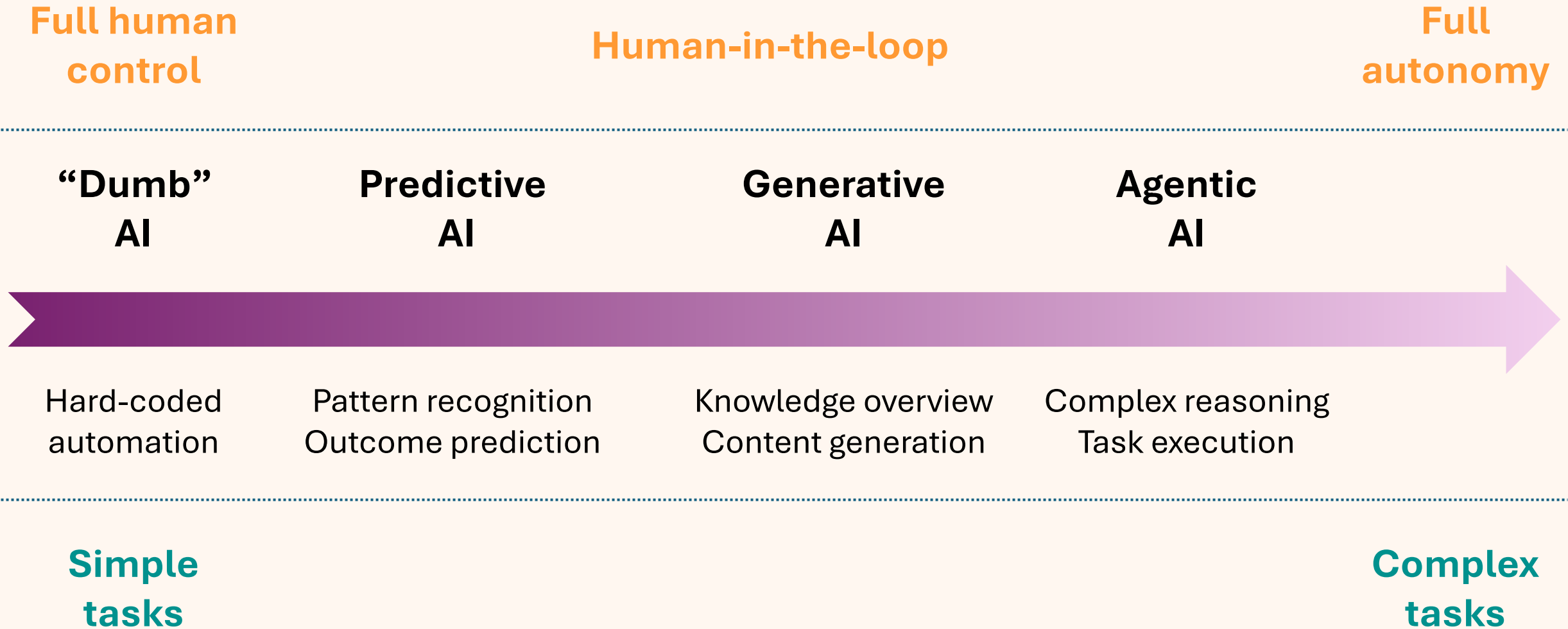
Knowledge overview
Content generation

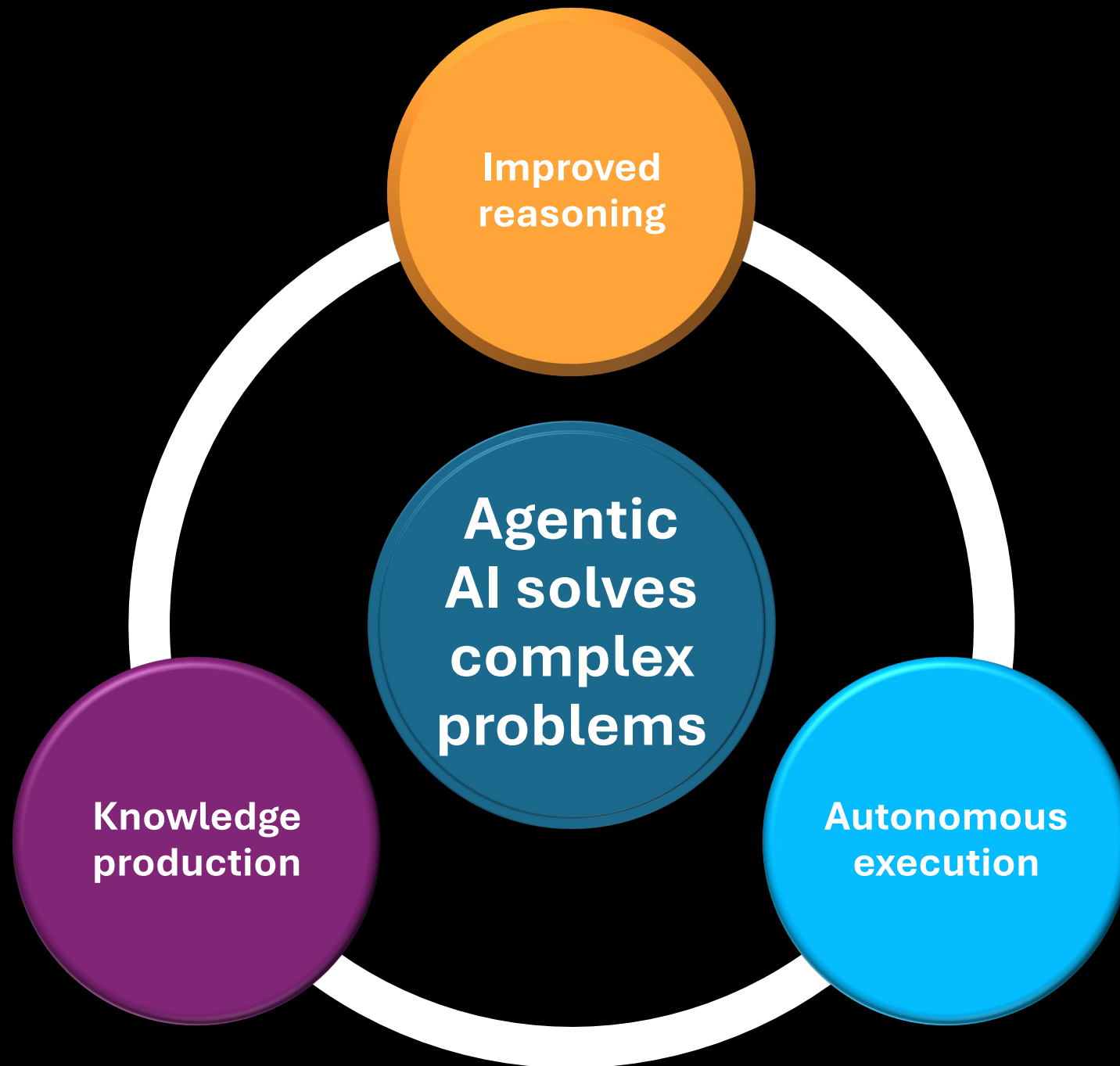
Complex reasoning
Task execution

**Simple
tasks**

**Complex
tasks**

AI has evolved in both capabilities and autonomy





Improved
reasoning

Generative AI

Prompt



Output

Improved
reasoning

Generative AI

Prompt
↓
Output

Agentic AI

Prompt
↓
Task decomposition
↓
Context collection
↓
Expert specialization
↓
“Chain of Thought” reasoning
↓
Multi-agent collaboration
↓
Review

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Review

**Iterative
refinement**



Improved
reasoning

Generative AI

Prompt
↓
Output

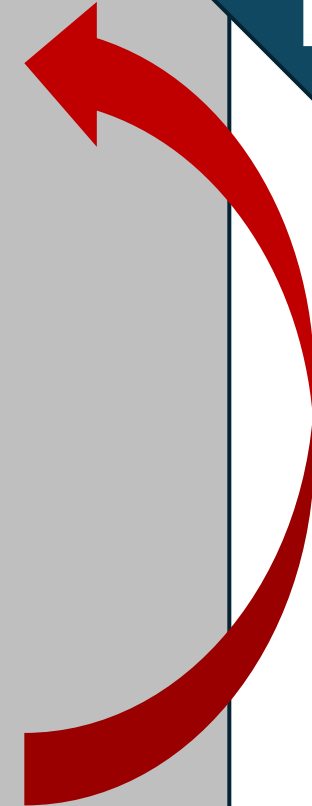
Agentic AI

Prompt
↓
Task decomposition
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Expert specialization
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Memory

**Iterative
refinement**



Improved
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Generative AI

Prompt
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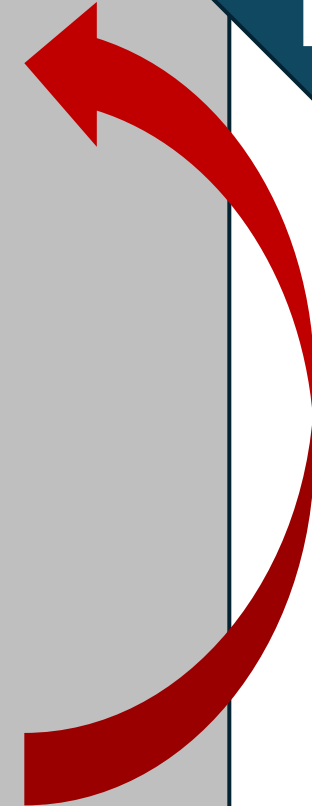
Agentic AI

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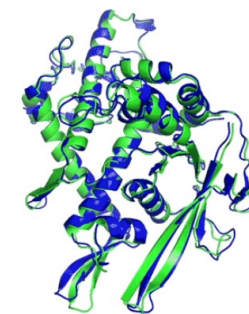
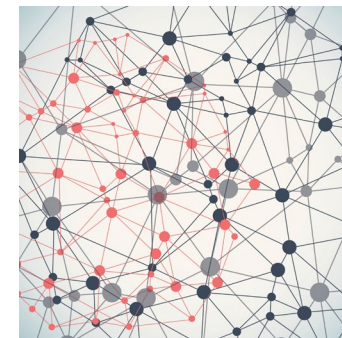
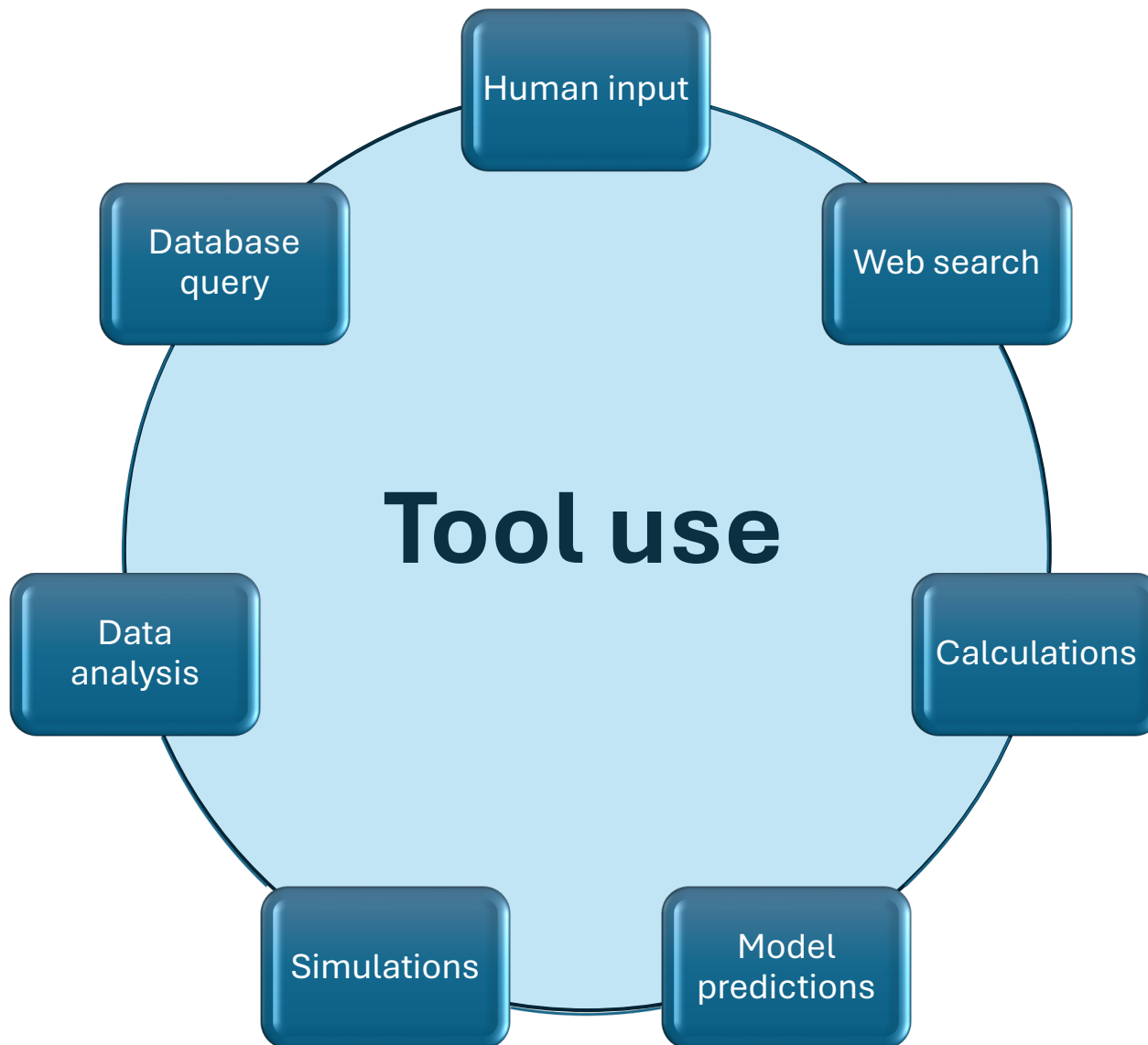
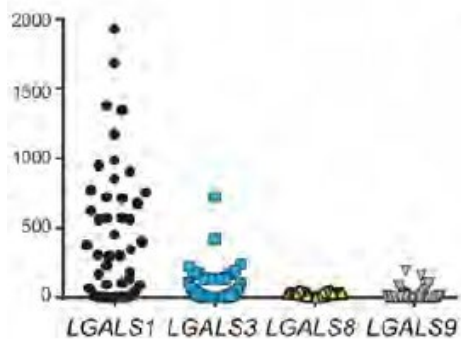
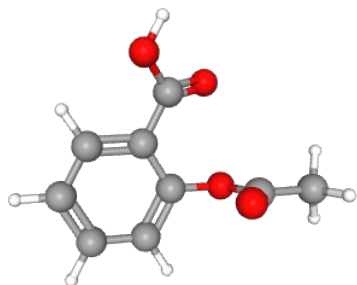


Memory

Iterative
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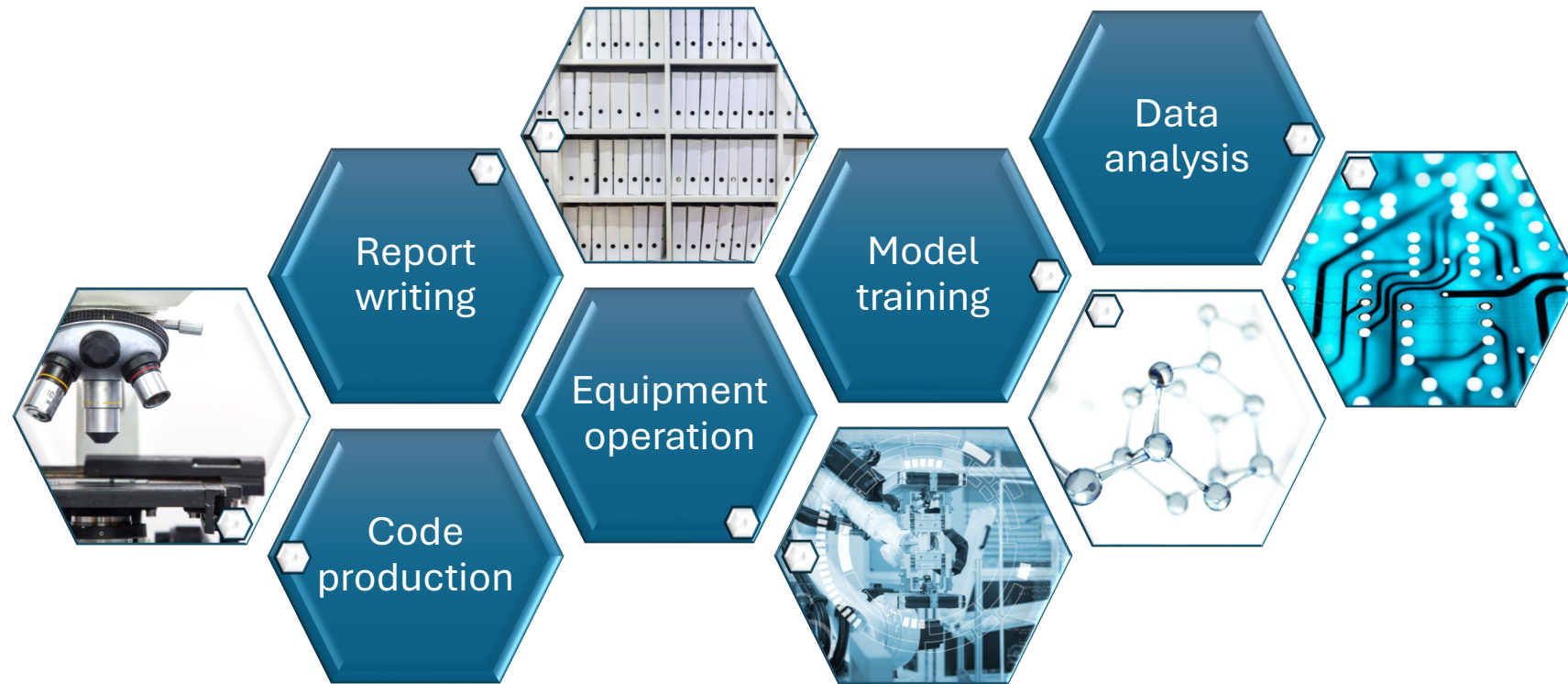


Knowledge
production



T1037 / 6vr4
90.7 GDT
(RNA polymerase domain)

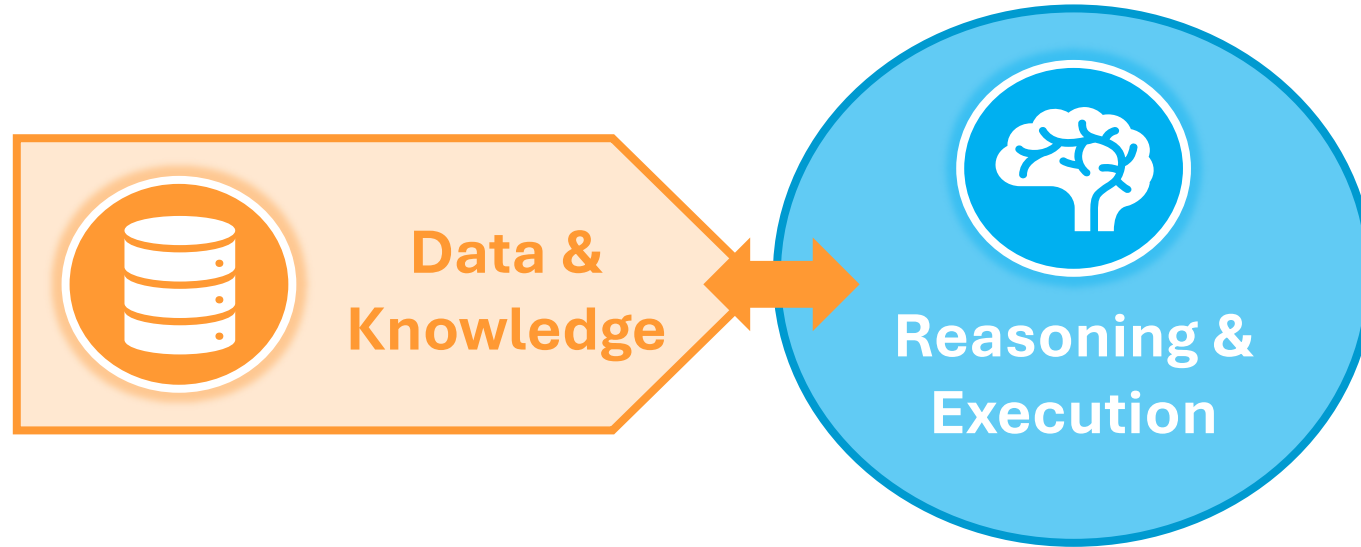
Autonomous execution



→ Execution of complex and long-running processes with minimal human input



**Reasoning &
Execution**







New Gem

Save

Describe your Gem and explain what it does

Instructions ⓘ

Example: You are a horticulturist with a background in natural lawns and native plants, and you help people plan gardens. Take into account location, weather, and what plants are



Default tool ⓘ

No default tool ▾

Knowledge ⓘ

Add files for your Gem to reference

Preview

To preview your Gem start by giving it a name

Ask Gemini

+

Fast ▾



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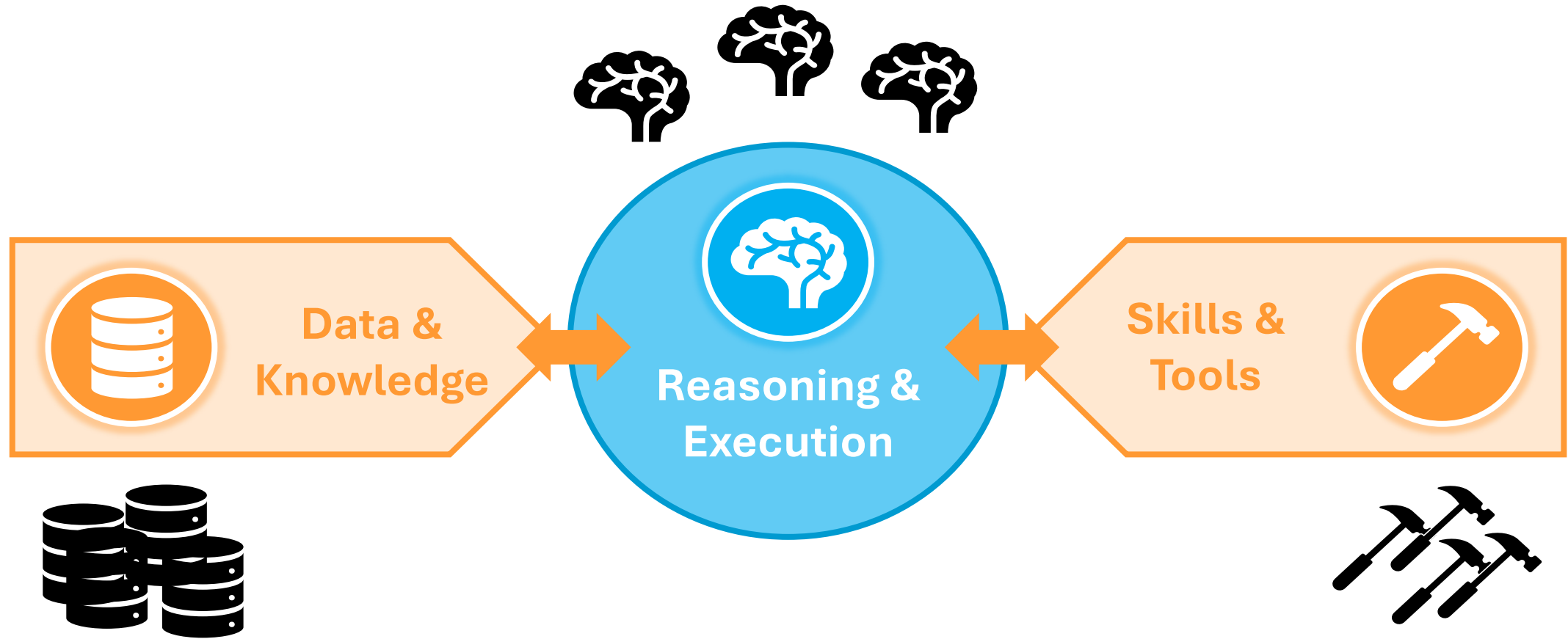
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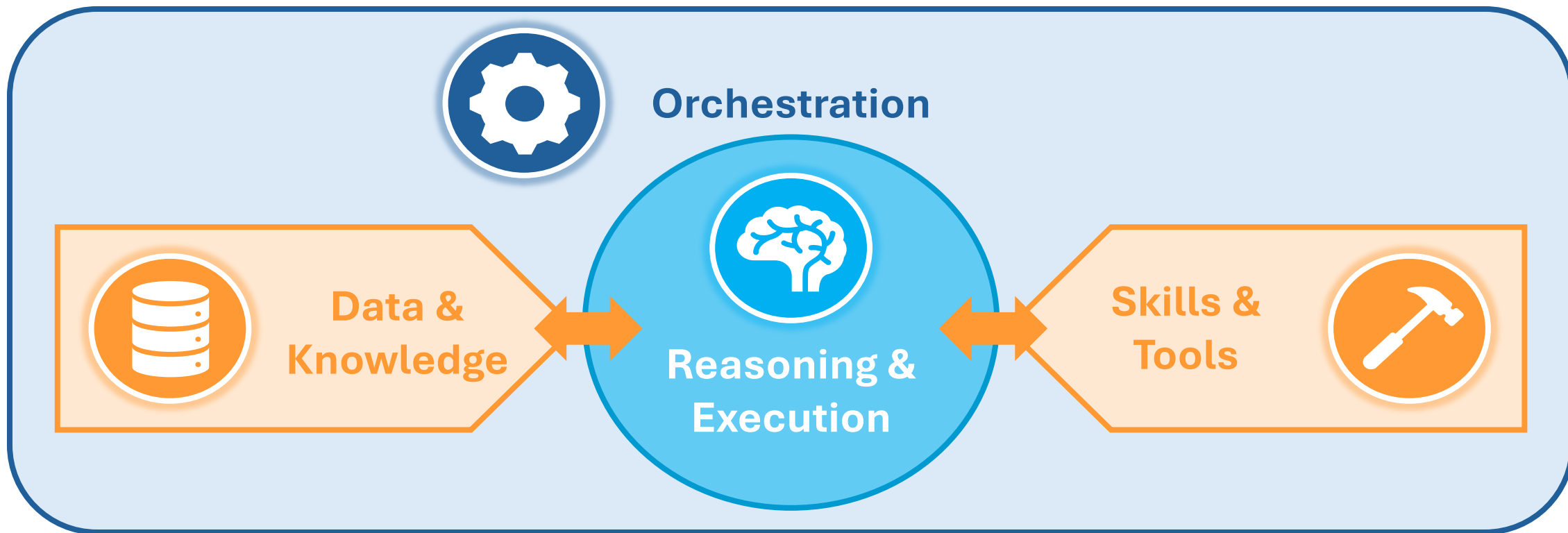
Preview

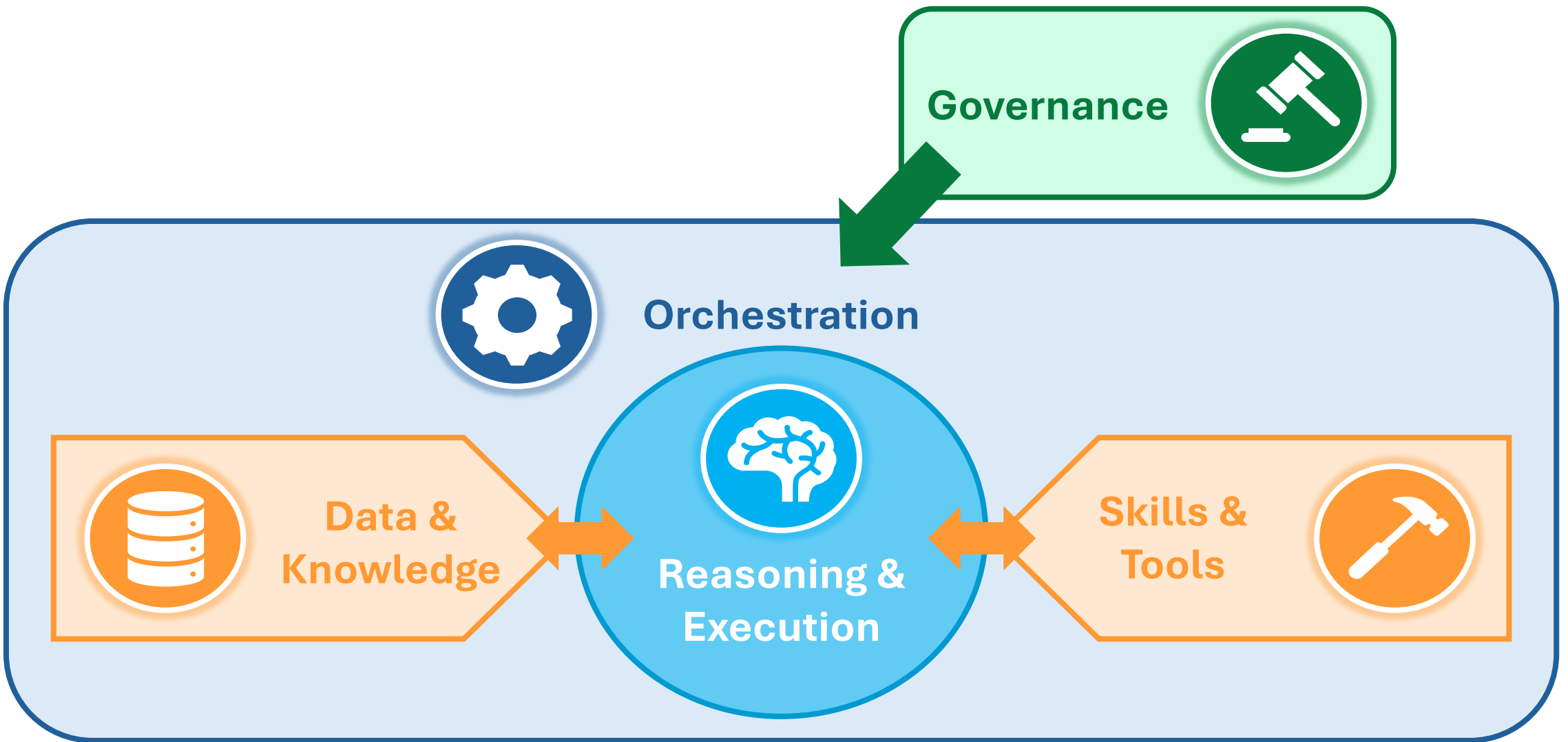
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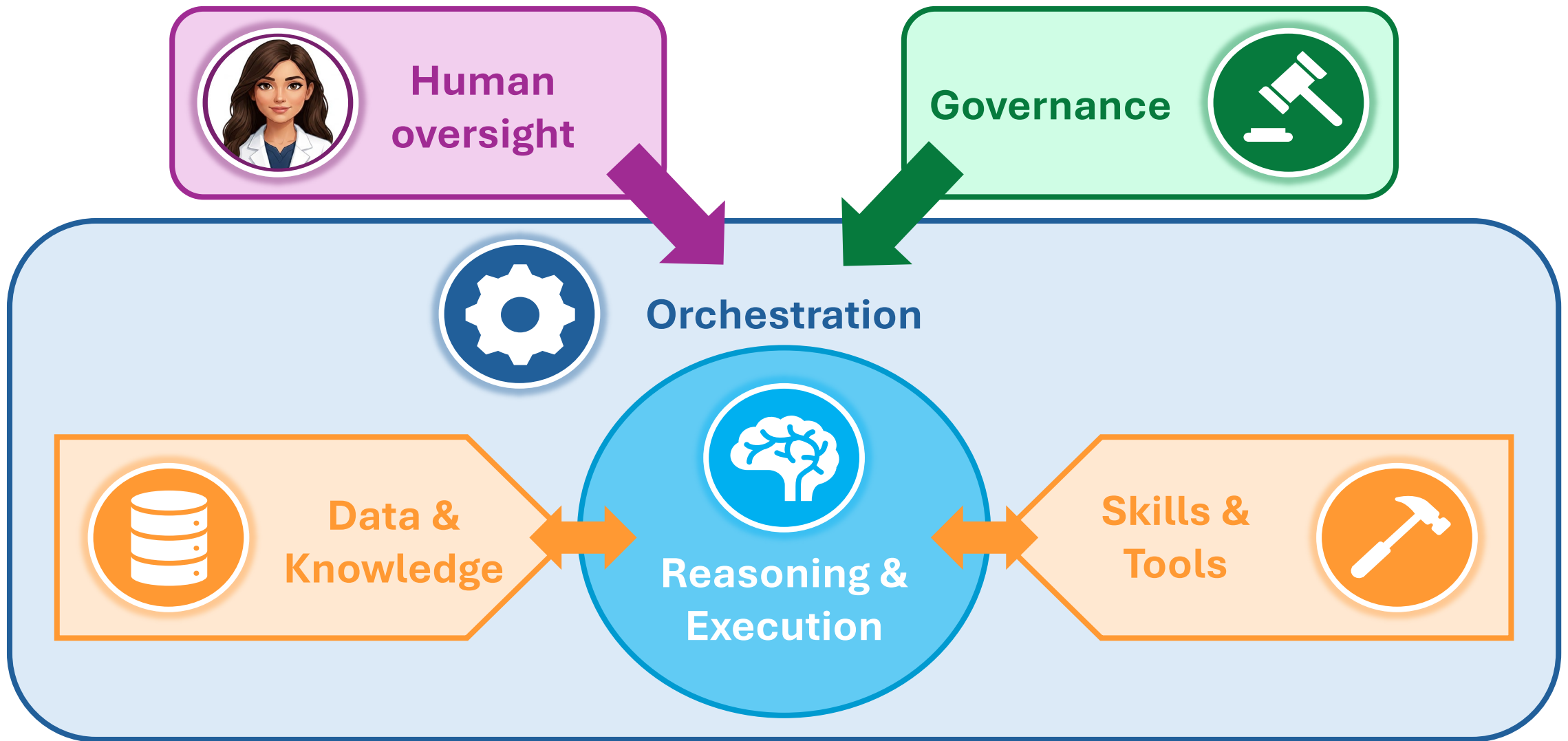
Ask Gemini

+ Fast ▾











**Human
oversight**

Governance



Orchestration



**Data &
Knowledge**



**Reasoning &
Execution**



**Skills &
Tools**



QC & Audits

Orchestration provides multi-agent coordination and operational control



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```
{  
  "workflow_id": "123",  
  "current_step": "toxicity_prediction",  
  "status": "running",  
  "retry_count": 1  
}
```

Coordinates agents, skills, tools and infrastructure

Manages workflows

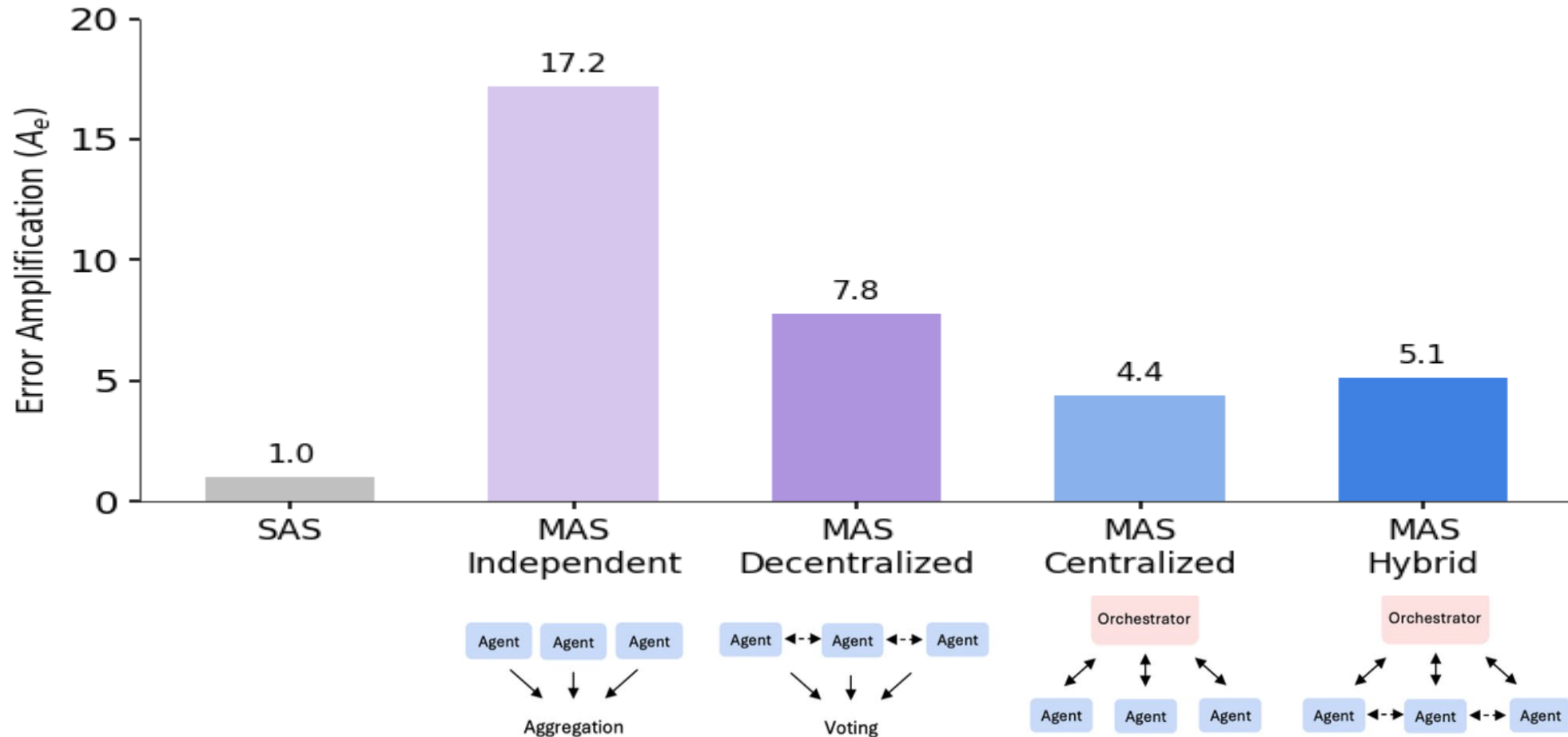
Maintains persistent state and memory

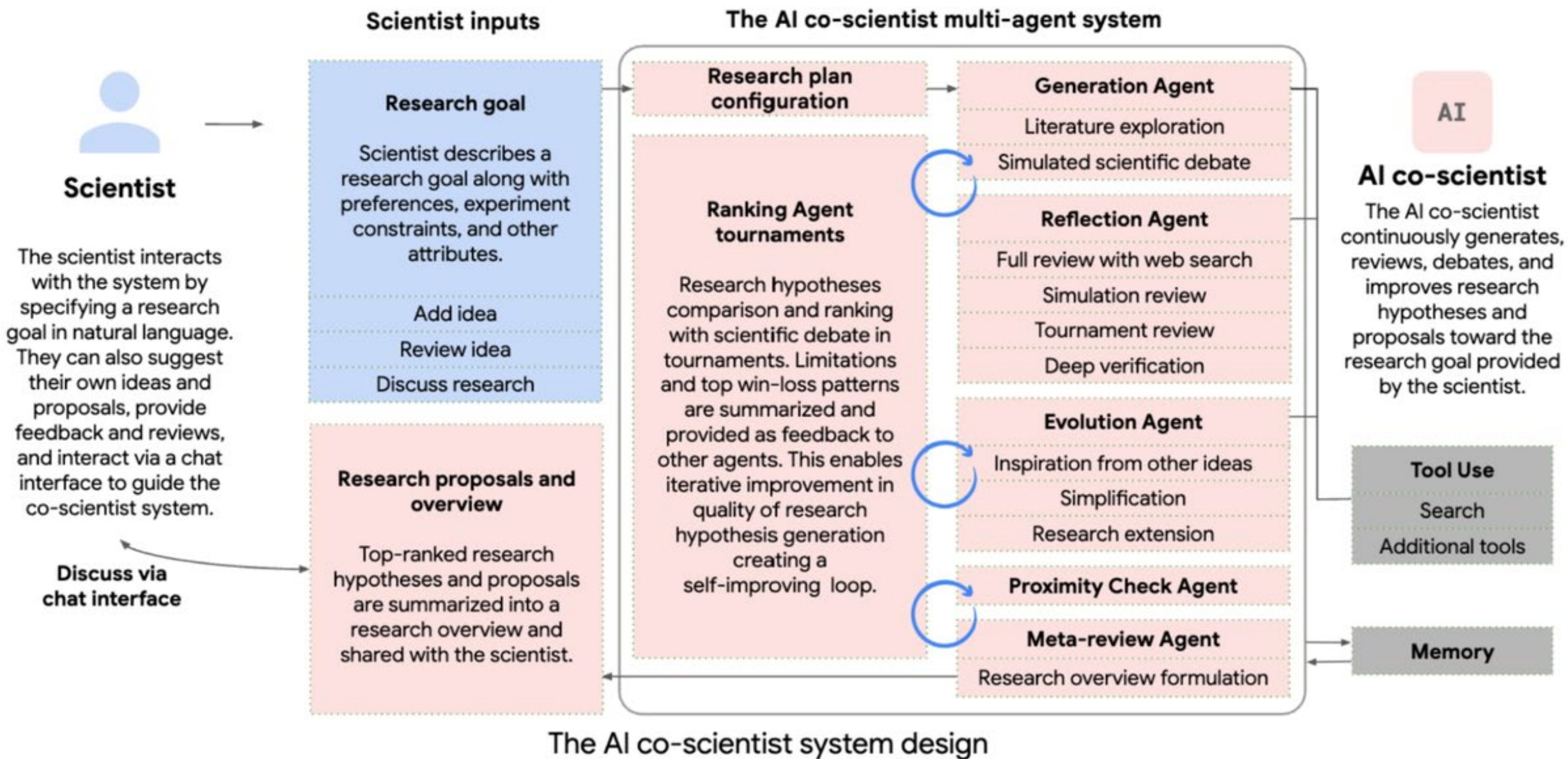
Handles failures, retries and recovery

Implements human oversight

Enforces governance guardrails

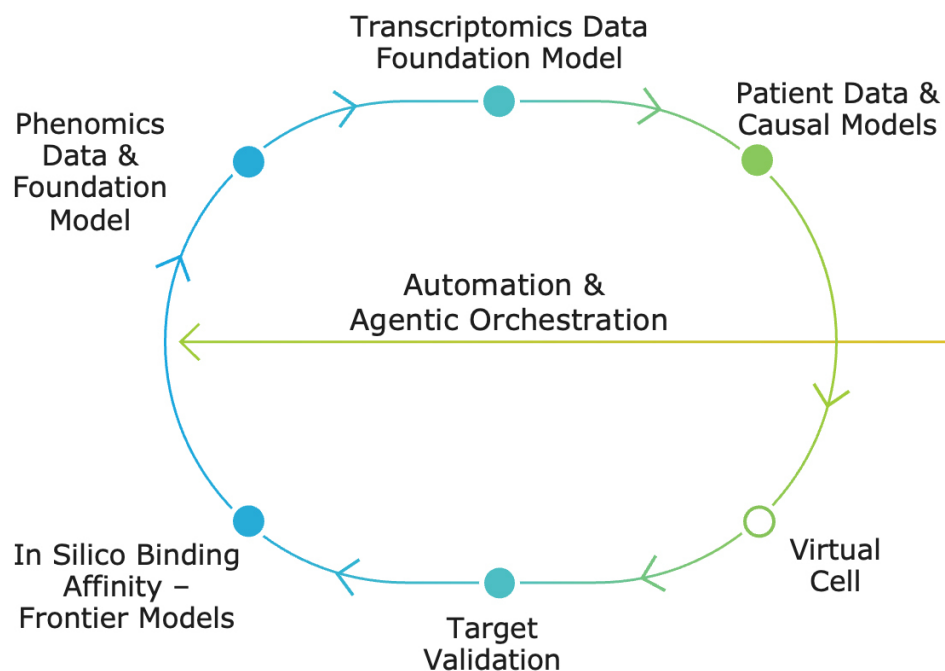
Lack of robust orchestration can lead to error amplification



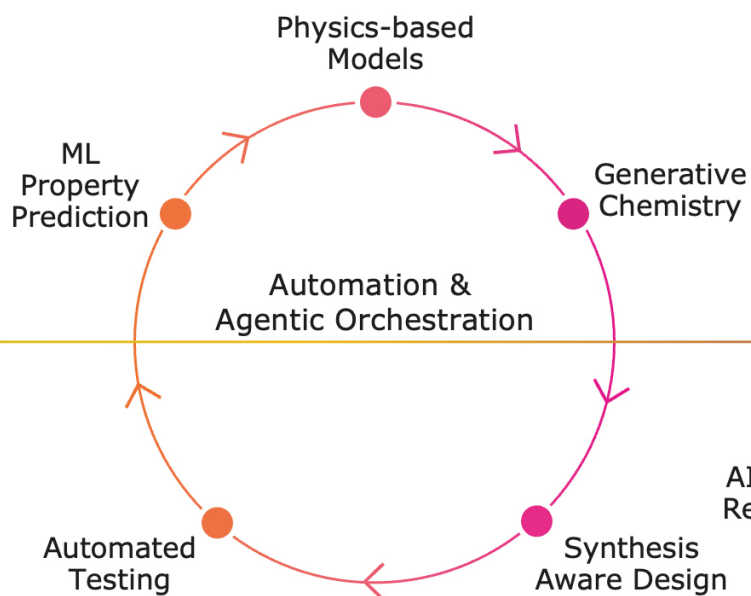


Agentic AI-native R & D

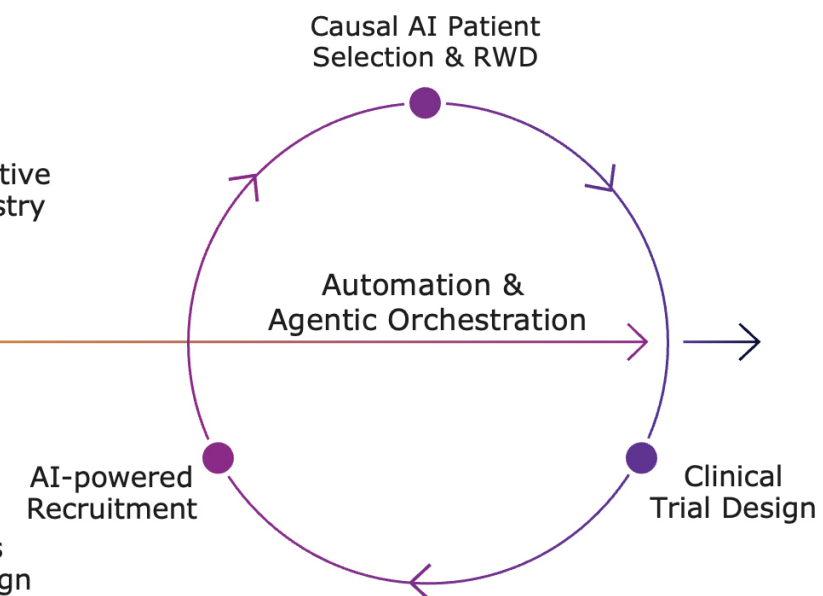
Biology to Insight



Insight to Molecule



Molecule to Patient



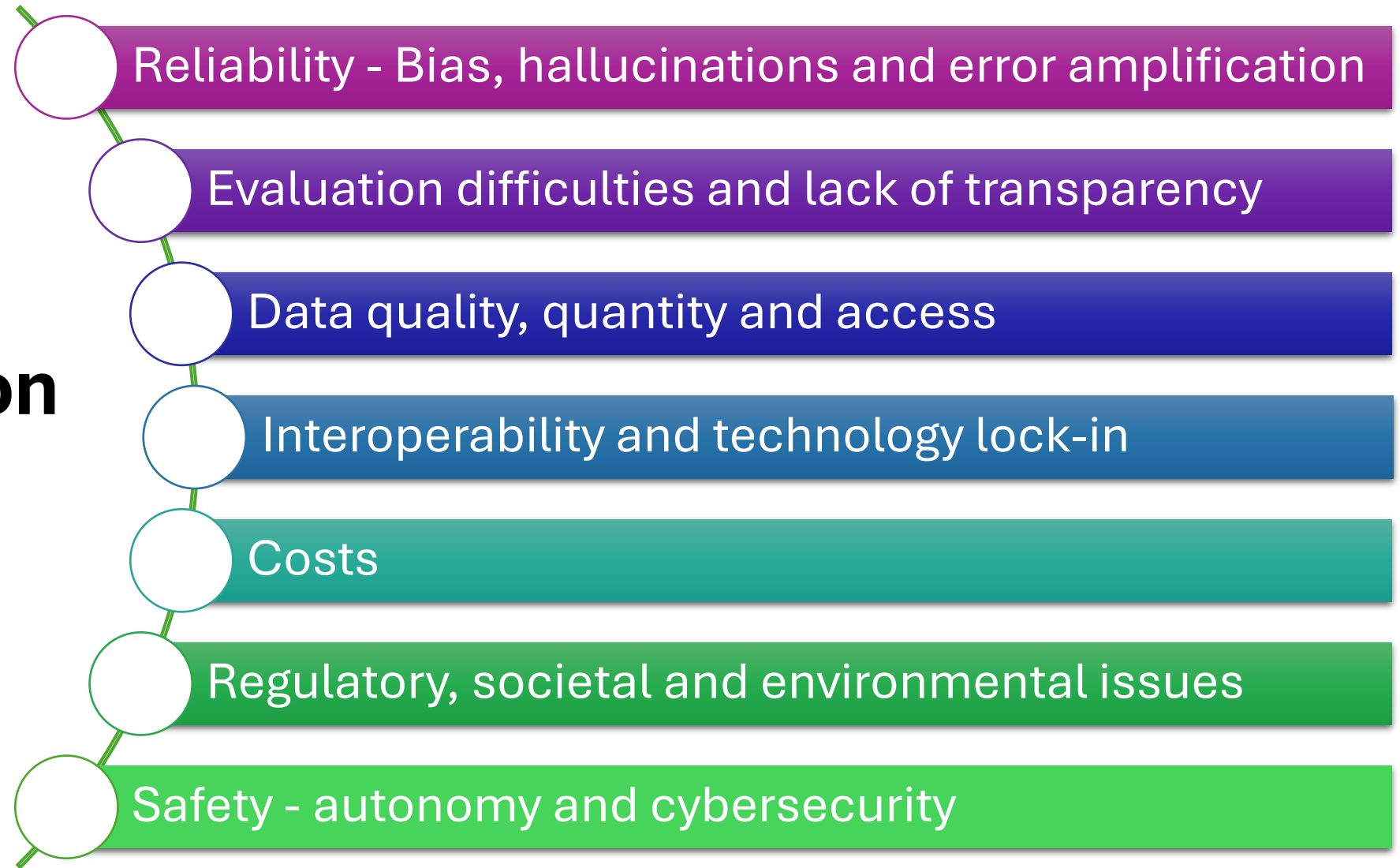
Agentic AI is a general-purpose technology

Intelligence gathering	Systems biology	Lab automation	Drug discovery	Clinical development	Approval/ Surveillance	Clinical practice
Literature review Database query Knowledge integration Patent review Competitive intelligence	Knowledge graph production Multi-omics integration Pathway analysis Biomarker discovery	Protocol generation Instrument operation Quality control Autonomous experiments Automatic documentation	Molecule generation Target/MoA identification SAR analysis Lead optimization ADMET prediction	Trial design Site selection/ orchestration Patient identification Patient monitoring Personalized engagement	Monitoring of regulatory requirements Compliance verification Document drafting Adverse event monitoring	Personalized prevention/ treatment Patient monitoring Diagnostic tools Treatment planning

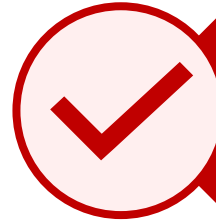
Business operations

Recruitment, Staff management, Supply chain operation, Infrastructure surveillance, ESG monitoring, financial reporting

Agentic AI implementation faces many challenges



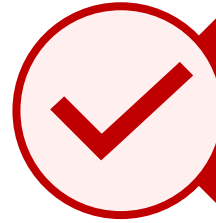
How to succeed with Agentic AI?



**Start simple and expand
iteratively**

**How to succeed
with Agentic AI?**

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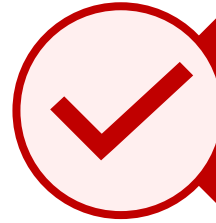


**Start simple and expand
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**Evaluate and control
vigorously**

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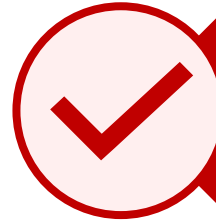


**Evaluate and control
vigorously**



Stay updated

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**Evaluate and control
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Stay updated



**Hire people with dual
expertise**

Know what to expect

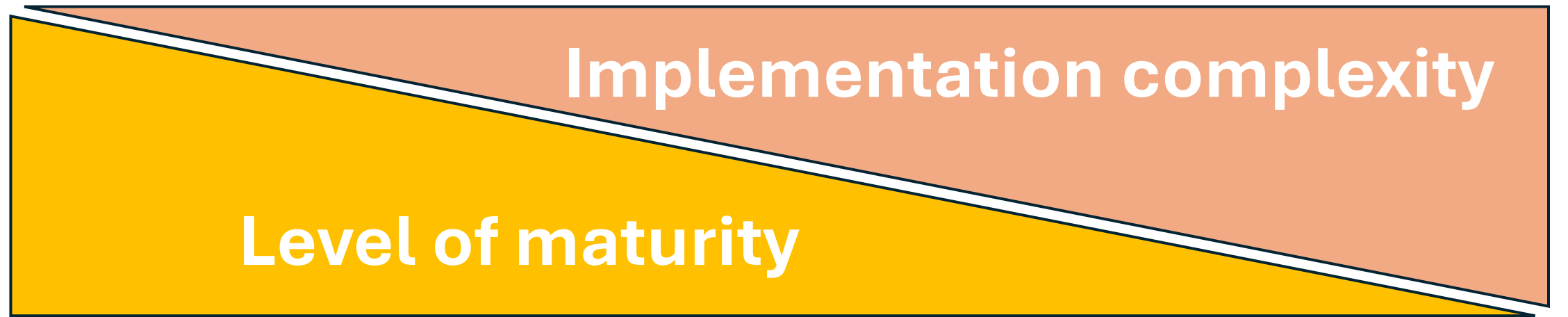
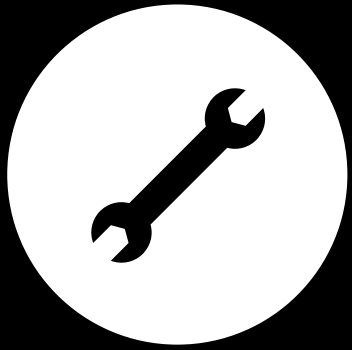


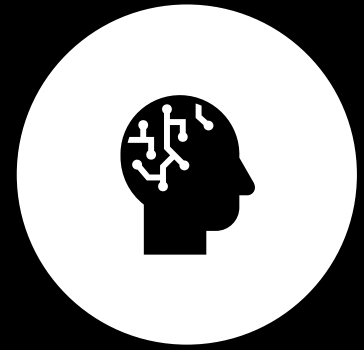
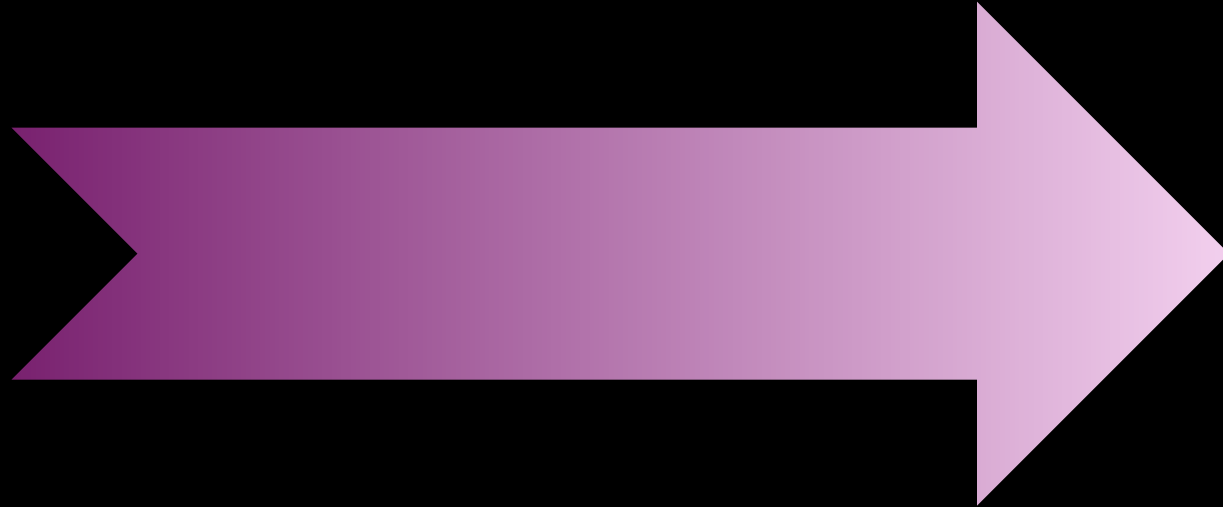
Image analysis
General chatbots
Remote monitoring

Structure prediction
Clinical decision support
Clinical trial optimization

Medical reasoning
Personalized medicine
Autonomous AI scientists



**AI as a
TOOL**



**AI as a
COLLEAGUE**

Take home message

- AI is moving towards complex agentic systems with increasing autonomy

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- Tool use provides a major boost to AI capabilities

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- Discriminative and generative AI remain important as building blocks in agentic AI systems and freestanding tools

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- Tool use provides a major boost to AI capabilities
- Robust orchestration is a key success factor
- Discriminative and generative AI remain important as building blocks in agentic AI systems and freestanding tools
- Lab/clinic integration is still in its infancy